Model comparisons across datasets and metrics

Dice

HCC-TACE-Seg

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	+0.297*	+0.478*	+0.024	+0.158*	-0.024	
DP	+0.321*	+0.502*	+0.048*	+0.182*		
TSLL	+0.139*	+0.320*	-0.134*			
TSLT	+0.273*	+0.454*				
AN	-0.181*					
COALA						

LiTS

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	+0.045*	+0.166*	+0.060*	+0.030*	-0.011	
DP	+0.056*	+0.178*	+0.071*	+0.042*		
TSLL	+0.015	+0.136*	+0.030*			
TSLT	-0.015	+0.107*				
AN	-0.122*					
COALA						

TCIA-CRLM

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	+0.026*	+0.109*	-0.045*	+0.070*	-0.004	
DP	+0.030*	+0.112*	-0.041*	+0.073*		
TSLL	-0.043*	+0.039*	-0.114*			
TSLT	+0.071*	+0.153*				
AN	-0.082*					
COALA						

WORC-CRLM

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	+0.060*	+0.205*	+0.037	+0.094*	+0.033*	
DP	+0.027*	+0.172*	+0.005	+0.061*		
TSLL	-0.034*	+0.111*	-0.057*			
TSLT	+0.022*	+0.167*				
AN	-0.145*					
COALA						

CAIRO5

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	+0.001	+0.146*	+0.049*	+0.008	+0.005	
DP	-0.004	+0.141*	+0.044*	+0.003		
TSLL	-0.006	+0.139*	+0.041*			
TSLT	-0.048*	+0.097*				
AN	-0.145*					
COALA						

Precision

HCC-TACE-Seg

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	+0.066	+0.005	+0.005	+0.011	+0.064*	
DP	+0.002	-0.058*	-0.059*	-0.053*		
TSLL	+0.055	-0.006	-0.006			
TSLT	+0.061	+0.000				
AN	+0.061*					
COALA						

LiTS

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	+0.033	-0.019	+0.077*	+0.034	-0.026*	
DP	+0.054*	+0.002	+0.098*	+0.055*		
TSLL	-0.001	-0.053*	+0.043*			
TSLT	-0.044	-0.096*				
AN	+0.052*					
COALA						

TCIA-CRLM

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	-0.026	-0.087*	-0.035*	+0.010	-0.028*	
DP	+0.003	-0.058*	-0.006	+0.039		
TSLL	-0.036	-0.097*	-0.045			
TSLT	+0.009	-0.052*				
AN	+0.061*					
COALA						

WORC-CRLM

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	-0.003	+0.003	+0.016	+0.007	-0.022	
DP	+0.021	+0.026	+0.039	+0.030		
TSLL	-0.009	-0.004	+0.009			
TSLT	-0.018	-0.013				
AN	-0.006					
COALA						

CAIRO5

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	+0.003	-0.033*	+0.049	+0.006	-0.006*	
DP	+0.010*	-0.027*	+0.056*	+0.012*		
TSLL	-0.003	-0.039*	+0.043			
TSLT	-0.046	-0.082*				
AN	+0.037*					
COALA						

Sensitivity

HCC-TACE-Seg

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	+0.351*	+0.519*	+0.025	+0.200*	-0.091*	
DP	+0.443*	+0.610*	+0.116*	+0.291*		
TSLL	+0.151*	+0.319*	-0.175*			
TSLT	+0.326*	+0.494*				
AN	-0.167*					
COALA						

LiTS

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	+0.045*	+0.244*	+0.041	+0.040*	+0.008*	
DP	+0.037	+0.236*	+0.033	+0.032*		
TSLL	+0.005	+0.204*	+0.001			
TSLT	+0.004	+0.202*				
AN	-0.198*					
COALA						

TCIA-CRLM

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	+0.074*	+0.219*	-0.026	+0.115*	+0.024*	
DP	+0.051*	+0.195*	-0.050*	+0.092*		
TSLL	-0.041*	+0.103*	-0.141*			
TSLT	+0.100*	+0.245*				
AN	-0.144*					
COALA						

WORC-CRLM

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	+0.076*	+0.228*	+0.046	+0.124*	+0.055*	
DP	+0.022	+0.173*	-0.009	+0.069*		
TSLL	-0.047*	+0.104*	-0.078*			
TSLT	+0.031*	+0.182*				
AN	-0.151*					
COALA						

CAIRO5

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	+0.002	+0.223*	+0.039*	+0.010	+0.012*	
DP	-0.011	+0.211*	+0.027	-0.002		
TSLL	-0.009	+0.213*	+0.028			
TSLT	-0.037	+0.184*				
AN	-0.221*					
COALA						

Detectability

HCC-TACE-Seg

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	+0.360*	+0.530*	+0.046	+0.107*	-0.029	
DP	+0.389*	+0.558*	+0.074*	+0.135*		
TSLL	+0.253*	+0.423*	-0.061			
TSLT	+0.315*	+0.484*				
AN	-0.170*					
COALA						

LiTS

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	+0.018	+0.051	+0.118*	-0.005	-0.015	
DP	+0.033	+0.066*	+0.133*	+0.009		
TSLL	+0.023	+0.057	+0.124*			
TSLT	-0.100*	-0.067*				
AN	-0.033					
COALA						

TCIA-CRLM

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	+0.007	+0.065	+0.023	+0.020	-0.006	
DP	+0.013	+0.071*	+0.029	+0.027		
TSLL	-0.013	+0.045	+0.002			
TSLT	-0.015	+0.042				
AN	-0.058*					
COALA						

WORC-CRLM

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	+0.009	+0.221*	+0.053	+0.039	-0.020	
DP	+0.029	+0.241*	+0.072	+0.059		
TSLL	-0.031	+0.182*	+0.013			
TSLT	-0.044	+0.169*				
AN	-0.213*					
COALA						

CAIRO5

	COALA	AN	TSLT	TSLL	**DP**	**SP**
SP	-0.014	+0.032	+0.110*	+0.003	+0.013	
DP	-0.027	+0.019	+0.097*	-0.010		
TSLL	-0.017	+0.030	+0.107*			
TSLT	-0.124*	-0.078				
AN	-0.047					
COALA						

Significantly higher Significantly lower Not significant

Voxel level: Dice, Precision, Sensitivity. Lesion level: Detectability (IoU ≥ 0.20).
Values: mean paired differences (row model − column model).
* Holm-adjusted Wilcoxon p < 0.05 (15 pairs per dataset × metric), with significant Holm-adjusted Friedman test.
SP–DP: valid SP–DP pairs; other pairs: cases valid across all six models.